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Artificial intelligence innovation, innovation niches and entrepreneurial performance: the curvilinear role of open innovation

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Abstract

Purpose – This study aims to examine how artificial intelligence (AI) innovation influences open innovation activities in new ventures, with a focus on the mediating role of innovation niches and the alignment between inbound and outbound open innovation in shaping entrepreneurial performance. Despite growing interest in AI-driven strategies, limited research explores these interactions in the context of emerging economies.

Design/methodology/approach – Using survey data from 329 high-tech venture founders in Vietnam, the study applies polynomial regression with response surface analysis to test the proposed hypotheses on the relationships between AI innovation, innovation niches, open innovation and entrepreneurial performance.

Findings – The results show that AI innovation enhances both inbound and outbound open innovation and strengthens innovation niches, which serve as key mediators. Additionally, the polynomial regression with response surface analysis reveals that entrepreneurial performance improves when inbound and outbound open innovation are aligned but declines under conditions of imbalance. This highlights the importance of congruence in open innovation strategies.

Practical implications – The findings offer actionable insights for entrepreneurs on leveraging AI and strategically balancing inbound and outbound open innovation, especially in emerging markets where resources and innovation capabilities are often limited.

Originality/value – This study contributes to the literature by integrating AI innovation, innovation niches and open innovation strategies into a unified framework. It offers novel empirical insights using PRRSA and responds to recent calls to understand how AI influences entrepreneurial strategies, particularly in resource-constrained and transitional economies like Vietnam.

Keywords AI innovation, Innovation niches, Open innovation, Entrepreneurial performance, New ventures

Paper type Research article

1. Introduction

The dynamic and competitive nature of today's markets has placed immense pressure on new ventures to innovate and achieve entrepreneurial success (Adomako *et al.*, 2024; Huang *et al.*, 2023). These ventures often operate with limited resources, requiring them to adopt innovative strategies to thrive in uncertain environments (Guo *et al.*, 2023). Artificial intelligence (AI) innovation has emerged as a transformative driver, enabling firms to enhance operational efficiency (Ku and Chen, 2024), adapt to volatile market conditions (Gama and Magistretti, 2025), and foster open innovation activities (Prasetyo *et al.*, 2025). From a socio-technical systems (STS) perspective, AI innovation can facilitate the integration of advanced technological tools and organizational strategies, encouraging knowledge acquisition, dissemination, and commercialization within entrepreneurial ecosystems (Jorzik *et al.*, 2024; Roundy, 2022).

However, while AI innovation is widely promoted for its potential to improve entrepreneurial performance, its actual outcomes remain inconsistent across different settings. Some studies argue that investments in AI yield limited returns when they are not aligned with organizational routines, strategic intent, or knowledge management practices (Füller *et al.*, 2022; Madanaguli *et al.*, 2024). Also, Huber *et al.* (2020) contend that small ventures may lack the absorptive capacity to translate AI capabilities into value or face challenges in coordinating AI with open innovation initiatives. These contradictions point to a critical knowledge gap: under what conditions does AI innovation actually drive performance

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in new ventures? This study responds by examining how AI interacts with innovation niches and open innovation dimensions to shape outcomes.

The relationship between innovation niches and open innovation offers significant insights into the functioning and dynamics of innovation ecosystems. Indeed, innovation niches reflect a firm's strategic positioning and resource advantages within the broader ecosystem, including its technological capabilities and market presence (Xie and Wang, 2021; Lopolito *et al.*, 2013). Firms that successfully occupy and leverage superior innovation niches are important in the ecosystem, as they are well-positioned to integrate external knowledge and optimize internal knowledge dissemination (Datta *et al.*, 2019). In this sense, innovation niches serve as relational hubs that facilitate resource flows and collaboration within open innovation ecosystems (Xie and Wang, 2021; Serrano-Ruiz *et al.*, 2025). Despite their relevance, the mediating role of innovation niches in linking AI innovation with open innovation strategies has received limited empirical attention. Exploring how firms strategically position themselves within these niches and align their resources with open innovation processes can deepen our understanding of the mechanisms that foster entrepreneurial performance and ecosystem sustainability.

Prior research highlights that open innovation enables firms to access diverse and novel knowledge from external sources, addressing market, environmental, and social challenges while providing critical resources for innovation processes (Wang *et al.*, 2023). This approach enhances firms' adaptability and influences their innovation performance, emphasizing the strategic value of open innovation in achieving organizational goals (Broekhuizen *et al.*, 2023; Sahoo *et al.*, 2024). The effectiveness of open innovation strategies depends significantly on a firm's ability to align and manage its knowledge acquisition and dissemination processes (Srisathan *et al.*, 2023), especially in resource-constrained environments (Šlapáková Losová and Dvouletý, 2024). Although inbound and outbound open innovation have been individually linked to firm performance (Wang *et al.*, 2015; Moretti and Biancardi, 2020; Abazi-Alili *et al.*, 2024), a little is known about the congruent and incongruent interaction between these two dimensions. Specifically, how their alignment or misalignment influences entrepreneurial outcomes remains unclear and is underexplored in the literature.

The empirical context of Vietnam offers a relevant setting to explore these questions. Vietnam has witnessed a surge in digital transformation and startup activity, supported by national AI development strategies and government-backed innovation hubs (Mai *et al.*, 2024). However, its entrepreneurial ecosystem is still developing, with startups often facing structural constraints, including limited R&D investment, informal knowledge flows, and institutional gaps (Đoàn Xuân, 2024). These characteristics make Vietnam a valuable case for investigating how new ventures navigate AI adoption and open innovation under contextual pressures. Studying Vietnam helps generate insights for emerging markets more broadly, where firms often pursue innovation amid uncertainty and policy-led ecosystems. To address these issues, this study investigates the mechanisms through which AI innovation, innovation niches, and the congruence of open innovation jointly influence entrepreneurial performance. More specifically, this study seeks to address the following research questions (RQs):

RQ1. How does AI innovation influence entrepreneurial performance through open innovation?

RQ2. How do innovation niches mediate the relationships between AI innovation, open innovation, and entrepreneurial performance?

RQ3. How does the congruence or incongruence between inbound and outbound open innovation affect entrepreneurial performance?

This study significantly contributes to the literature on AI innovation, open innovation, and entrepreneurial performance. First, it enhances understanding of how AI innovation drives strategic alignment and facilitates knowledge flows within entrepreneurial ecosystems. By identifying AI as a catalyst for inbound and outbound open innovation, this research

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addresses the mechanisms through which AI strengthens firms' knowledge management and innovation capabilities. Second, the study highlights the mediating role of innovation niches in linking AI innovation to open innovation strategies. This research offers valuable insights into optimizing open innovation processes by demonstrating how firms leverage their strategic positioning and resource advantages within innovation ecosystems. Third, this study employs an advanced analytical approach—Polynomial regression with response surface analysis (PRRSA)—which moves beyond traditional linear models to uncover more complex, nonlinear relationships between inbound and outbound open innovation. This method allows for a precise examination of the congruence (alignment) and incongruence (misalignment) between the two dimensions, providing nuanced insights into how these configurations influence entrepreneurial performance. Compared to previous linear-only studies, such as Sriram and Hungund (2022) and Bigliardi *et al.* (2020), PRRSA enables visualization of surface patterns and interaction effects that are otherwise masked in linear models, thus strengthening the study's methodological and theoretical originality. This finding contributes to the organizational fit literature, advancing the understanding of how congruence between innovation activities influences outcomes.

The paper is organized as follows: Section 2 develops the theory and hypotheses. Section 3 describes the research methodology. Section 4 presents the results, and Section 5 discusses the findings. Section 6 concludes with contributions, implications, limitations, and future research directions.

2. Theory and hypotheses

2.1 Theoretical foundation

The STS theory offers a robust conceptual framework for understanding how organizations can effectively integrate technological advancements, such as AI, into their social and operational structures to drive innovation and enhance performance. According to STS theory, organizations operate as complex systems made up of interconnected social and technical subsystems that must be harmonized to optimize outcomes (Sahoo *et al.*, 2024; Ku and Chen, 2024). The social subsystem encompasses human actors, their roles, and the cultural and interpersonal dynamics within the organization, while the technical subsystem includes the tools, processes, and technological infrastructure that facilitate organizational activities (Thomas, 2024). As a significant technological advancement, AI innovation revolutionizes the technical subsystem by introducing new approaches to processes, products, and services (Ku and Chen, 2024). These innovations are crucial in promoting open innovation—a paradigm that emphasizes the active exchange of ideas, knowledge, and technologies among both internal and external stakeholders (Broekhuizen *et al.*, 2023; Prasetyo *et al.*, 2025). From the STS perspective, the success of open innovation relies on creating an environment where AI-driven innovations are seamlessly integrated with an organization's social components (Holgersson *et al.*, 2024). This integration ensures that advancements in AI are effectively implemented and utilized to foster collaboration, stimulate creativity, and enhance decision-making processes.

However, socio-technical integration is often an iterative, non-linear process. For example, introducing an advanced AI tool (technical subsystem) can disrupt established routines or encounter employee resistance in the social subsystem, illustrating potential conflicts when technology outpaces the organization's readiness (Makarius *et al.*, 2020). Conversely, a strong learning culture and proper training (social factors) can synergistically harness new AI tools (technical factors), creating positive feedback loops of continuous innovation and learning (Thomas, 2024). In resource-constrained new ventures – such as those in Vietnam's emerging high-tech sector – these interactions are highly dynamic and adaptive; changes in one subsystem necessitate adjustments in the other to maintain alignment amidst limited resources and evolving practices.

The concept of open innovation, initially proposed by Chesbrough (2003), represents a shift in how firms manage innovation. Rather than relying solely on internal resources, this

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approach encourages companies to incorporate external knowledge and share internal innovations through multiple channels. Open innovation promotes the integration of diverse idea sources—both internal and external—into cohesive innovation systems, including platforms and frameworks that support technology development and commercialization. Bogers *et al.* (2018) argue that Open innovation practices rely on business models to outline the necessary structures and frameworks for innovation systems. These models leverage a combination of internal and external knowledge sources to generate value and simultaneously establish internal mechanisms that enable firms to capture a share of the value they help create. In the context of this study, STS theory serves as a foundation for examining how AI innovation influences two key dimensions of open innovation: inbound open innovation and outbound open innovation. Inbound open innovation refers to the process of absorbing external knowledge through business networks and collaborations with other external entities (Pinarello *et al.*, 2022). In contrast, outbound open innovation involves the transfer of internal knowledge or technology to other technological fields, firms, or industries (Wang *et al.*, 2023). These two types of open innovation represent a bidirectional process of knowledge exchange, with inbound open innovation emphasizing the inflow of external knowledge and outbound open innovation focusing on the outflow of internal knowledge (Arias-Pérez and Huynh, 2023; Çera *et al.*, 2024). The success of these activities relies on the interaction between the organization's social and technical subsystems. For example, AI-driven innovations can support the identification and evaluation of opportunities for external knowledge acquisition while fostering a collaborative culture that ensures these innovations are effectively utilized to generate value (Broekhuizen *et al.*, 2023).

An innovation niche, rooted in ecological concepts, represents a resource space for innovation and the productive capacities that emerge from interactions and competition within an organizational ecosystem (Xie and Wang, 2021). Within an open innovation ecosystem, a firm's niche includes its market position, resource availability—such as human, production, and innovation resources—technological status, and capabilities in operational, marketing, and organizational innovation (Heaton and Min, 2025). In this context, innovation niches serve as mediating constructs that link AI innovation to both inbound and outbound open innovation. Organizations can expand their resource spaces and enhance their innovative capacities by promoting AI-driven advancements that align with their strategic objectives (Broekhuizen *et al.*, 2023). This alignment, bolstered by a collaborative culture and adaptive organizational structures, allows firms to effectively utilize their innovation niches to enhance inbound knowledge absorption and outbound knowledge dissemination (Xie and Wang, 2021). Moreover, the interactions between inbound and outbound open innovation—whether harmonious or discordant—highlight the necessity of balancing these dimensions. A well-aligned innovation niche facilitates this balance and optimizes a firm's ability to leverage open innovation for improved entrepreneurial performance.

By framing this investigation within STS theory, the study emphasizes the strategic importance of integrating AI technologies into the socio-technical fabric of organizations. This alignment not only supports open innovation initiatives but also ensures that these initiatives contribute to sustained entrepreneurial success in dynamic and competitive (Ku and Chen, 2024; Sahoo *et al.*, 2024). The theoretical insights derived from STS highlight the interconnectedness of social and technical factors, offering a comprehensive perspective on the transformative potential of AI in driving innovation and performance.

2.2 AI innovation and innovation niches

AI innovation, by fostering the development of advanced technologies and processes, equips organizations with the ability to enhance their competitive positions within open innovation ecosystems (Ku and Chen, 2024). Previous studies have indicated that innovation niches—as resource spaces shaped by inter-organizational interactions—are heavily influenced by a firm's ability to adapt and innovate in response to the shifting demands of the market (Lopolito *et al.*,

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2011, 2013; Xie and Wang, 2021). AI innovation enables organizations to identify and leverage unique technological opportunities, optimize resource allocation, and respond effectively to competitive pressures (Broekhuizen *et al.*, 2023; Füller *et al.*, 2022). These advancements expand the firm's resource space, strengthening its innovation niche and enhancing its capacity to compete and collaborate within its ecosystem. Nevertheless, the extent to which AI innovation actually strengthens a venture's innovation niche may depend on the firm's strategic alignment and absorptive capacity. Some studies suggest that investments in AI yield limited competitive benefits when they are not aligned with organizational routines or knowledge management practices, or when firms lack the capacity to translate AI capabilities into value (Olan *et al.*, 2022; Kemp, 2024). The following hypothesis is formulated:

H1. AI innovation positively affects innovation niches.

2.3 AI innovation and open innovation

AI innovation supports organizations in acquiring and integrating external knowledge by providing advanced tools for data analysis, pattern recognition, and decision-making (Elahi *et al.*, 2023; Gama and Magistretti, 2025). These capabilities enable firms to effectively identify relevant external knowledge sources and streamline the process of collaboration with external entities (Sahoo *et al.*, 2024). Existing studies indicate that the efficiency of knowledge absorption and utilization is significantly improved when firms leverage innovative technologies to process and evaluate external inputs (Bashir, 2024; Ouyang *et al.*, 2023). AI-driven insights allow firms to target specific opportunities and partnerships, which can increase the effectiveness of inbound open innovation activities. However, simply deploying AI tools does not guarantee improved inbound knowledge absorption (Arndt *et al.*, 2023). Organizations lacking sufficient learning routines or absorptive capacity may struggle to convert AI-generated insights into valuable external collaborations (Bag *et al.*, 2023), meaning that the presence of AI must be coupled with organizational readiness to fully realize inbound open innovation benefits.

H2. AI innovation positively affects inbound open innovation.

AI innovation might facilitate the external dissemination of a firm's internal knowledge by enabling efficient knowledge transfer and technological sharing. With AI, organizations can analyze and package their intellectual assets in ways that maximize their relevance and applicability to external stakeholders (Füller *et al.*, 2022; Kaur *et al.*, 2023). Moreover, AI-driven platforms and tools enhance the scalability and accessibility of outbound innovation activities by automating routine processes and optimizing resource deployment. Prior studies highlight that the strategic use of technology fosters inter-firm collaboration and accelerates the diffusion of innovations across industries and markets (Redmond, 2004; Palm, 2022; Massa *et al.*, 2023). Consequently, AI innovation supports firms in strengthening their outbound open innovation by enabling the effective transfer and commercialization of internal knowledge. On the other hand, if an organization lacks a supportive culture or clear strategy for sharing its knowledge, the mere availability of AI technology may not significantly boost outbound open innovation (Lam *et al.*, 2021). Small ventures, in particular, can face challenges in coordinating AI initiatives with open innovation efforts (Del Vecchio *et al.*, 2018), indicating that technology-driven outbound activities will falter unless employees and processes are geared toward external collaboration.

H3. AI innovation positively affects outbound open innovation.

2.4 Innovation niches and open innovation

Innovation niches provide firms with enhanced access to resources and capabilities, effectively positioning them to absorb external knowledge through inbound open

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innovation (Xie and Wang, 2021). Firms with well-developed innovation niches—characterized by competitive market positions, advanced technological capabilities, and resource richness—are better equipped to engage in collaborative networks and attract external knowledge sources (Lopolito *et al.*, 2013). Research indicates that firms with a robust innovation niche are likelier to foster trust and credibility within their ecosystems, facilitating smoother knowledge acquisition and integration (Bienkowska *et al.*, 2010; Duong *et al.*, 2022). As such, innovation niches are a foundation for enhancing a firm's ability to identify, acquire, and utilize external knowledge effectively. It is important to note, however, that a strong niche does not automatically translate into openness to external ideas. Organizations that are very dominant or self-sufficient in their niche may develop a "not-invented-here" bias – an internal resistance against knowledge originating from outside the firm (Hussinger and Wastyn, 2016). Such insularity can temper the positive effect of an innovation niche on inbound open innovation if firms become complacent or dismissive of external knowledge.

H4. Innovation niches positively affect inbound open innovation.

Innovation niches not only support the acquisition of external knowledge but also enhance a firm's ability to disseminate its internal knowledge through outbound open innovation. Firms operating within well-established niches possess the necessary resources, networks, and technological expertise to share their innovations effectively with external stakeholders (Caiafa *et al.*, 2023). Xie and Wang (2021) also suggest that firms in strong innovation niches are perceived as valuable contributors within their ecosystems, increasing the likelihood that their knowledge assets are adopted and utilized by external entities. Furthermore, these firms can better align their knowledge-sharing strategies with market demands, which can strengthen their outbound innovation capabilities. Still, even resource-advantaged firms may hesitate to share internal innovations freely. Concerns about losing competitive advantage or intellectual property can make firms cautious in outbound open innovation (Lee *et al.*, 2010). If a venture perceives significant risks or costs in external knowledge transfer, its robust niche might not fully translate into greater outbound openness.

H5. Innovation niches positively affect outbound open innovation.

Innovation niches act as a crucial intermediary through which the effects of AI innovation on open innovation are assumed. AI innovation enhances a firm's capabilities and resources, enabling the development of a strong innovation niche. This enhanced niche, in turn, facilitates both inbound and outbound open innovation by providing a platform for effective knowledge acquisition and dissemination. The mediating role of innovation niches is supported by evidence that resource-rich and strategically positioned firms are more successful in leveraging technological advancements to engage in open innovation activities (Xie and Wang, 2021; Noviaristanti *et al.*, 2024). Thus, the influence of AI innovation on open innovation is amplified when it contributes to establishing and expanding a firm's innovation niche, which bridges technological advancements with knowledge transfer processes.

H6. Innovation niches mediate the relationships between AI innovation and open innovation (inbound and outbound open innovation).

2.5 Open innovation and entrepreneurial performance

Inbound open innovation facilitates entrepreneurial performance by enabling firms to acquire diverse external knowledge and resources, which are critical for innovation and market responsiveness (Liao *et al.*, 2020; Abazi-Alili *et al.*, 2024). This process allows firms to expand their knowledge base, reduce uncertainty in innovation activities, and respond effectively to emerging market demands. Moretti and Biancardi (2020) highlight that integrating external knowledge can help firms identify new opportunities, develop innovative solutions, and enhance their competitive positioning. By fostering collaboration with external

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stakeholders, firms can enhance their innovation capabilities, leading to improved entrepreneurial outcomes such as profitability, growth, and market share. Yet not all studies find uniformly positive effects of inbound openness. If firms source too broadly or rapidly from external domains, they may encounter information overload and difficulty assimilating the knowledge, resulting in diminishing returns on innovation efforts. Moreover, without adequate absorptive capacity and internal processes to utilize external inputs, the benefits of inbound open innovation might not materialize fully (Wang *et al.*, 2020). Moreover, inbound open innovation can serve as a critical mediating mechanism by which the benefits of AI innovation and innovation niches are translated into entrepreneurial performance. AI innovation enables firms to identify and absorb valuable external knowledge (Gama and Magistretti, 2025), while innovation niches provide the structural and resource-based foundations necessary for effective knowledge integration (Lopolito *et al.*, 2013). Firms thus can leverage these technological and resource advantages through inbound open innovation to enhance their innovative capabilities and performance outcomes.

H7. Inbound open innovation positively affects entrepreneurial performance.

H8. Inbound open innovation mediates the relationships between AI innovation, innovation niches, and entrepreneurial performance.

Firms can build strategic alliances, access complementary capabilities, and expand their presence in new markets by transferring resources, such as internal knowledge and technologies, to external stakeholders. Knowledge dissemination enhances a firm's ability to generate additional revenue streams, foster innovative collaborations, and strengthen its competitive position within the industry (Ode and Ayavoo, 2020; Audretsch *et al.*, 2023). Outbound innovation activities thus also provide firms access to new markets and networks, facilitating the diversification of revenue streams and improved resource allocation. Furthermore, effective management of external collaborations reduces the uncertainties associated with knowledge transfer and increases the success rate of commercialization efforts (Adomako and Nguyen, 2024; Sunnemark *et al.*, 2024). However, if firms engage heavily in outbound innovation without complementary inbound learning or without safeguards, they risk dissipating their knowledge advantages. Unchecked outbound knowledge flow can lead to intellectual property leakage or coordination difficulties that undermine the expected performance gains (Alberti and Pizzurno, 2017). In other words, sharing innovations externally must be accompanied by value capture mechanisms; otherwise, the positive impact of outbound open innovation on performance may be limited. Moreover, outbound open innovation is a mediating mechanism that links AI innovation and innovation niches to entrepreneurial performance by enabling internal capabilities and resource commercialization. AI innovation enhances the generation of high-value knowledge, while innovation niches provide the structural and resource-based conditions necessary for effective external collaboration (Olan *et al.*, 2022). Firms also can transfer and apply their technological and organizational advancements to external environments through outbound open innovation, fostering collaborative development and knowledge exchange (Srisathan *et al.*, 2023). By bridging internal innovation capabilities with external opportunities, outbound open innovation amplifies the synergistic effects of AI innovation and innovation niches on entrepreneurial performance.

H9. Outbound open innovation positively affects entrepreneurial performance.

H10. Outbound open innovation mediates the relationships between AI innovation, innovation niches, and entrepreneurial performance.

While both inbound and outbound open innovation are crucial for firm performance (Moretti and Biancardi, 2020; Wang *et al.*, 2015; Liao *et al.*, 2020), their alignment is expected to have synergistic effects, especially when both are pursued at high levels. A balanced increase in these two dimensions enables firms to fully leverage external knowledge while effectively disseminating their internal innovations, creating a dynamic and reciprocal knowledge

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exchange cycle. Sahoo *et al.* (2024) suggests that inbound open innovation enhances the firm's knowledge base, allowing for more informed decision-making and innovative solutions, while outbound open innovation builds external relationships and commercializes internal knowledge. When inbound and outbound open innovation are congruently high, firms can simultaneously acquire complementary knowledge and establish collaborative networks, leading to enhanced creativity, innovation output, and competitive positioning (Wang *et al.*, 2023). Consequently, coupled open innovation, which balances inbound and outbound activities, delivers superior innovation performance by optimizing resources, enhancing partnerships, and driving entrepreneurial performance.

H11. A congruent increase of inbound and outbound open innovation would enhance entrepreneurial performance.

An imbalance between inbound and outbound open innovation can undermine entrepreneurial performance, where one is pursued at a significantly higher level than the other. Firms may encounter information overload and inefficient resource allocation challenges if inbound open innovation exceeds outbound efforts. Excessive reliance on external knowledge without adequate outbound innovation to share and commercialize internal knowledge leads to underutilization of resources and missed opportunities for value creation (Masucci *et al.*, 2020; Wang *et al.*, 2023). Conversely, when outbound open innovation surpasses inbound efforts, firms may lack the external knowledge inputs needed to sustain their innovation activities. Without sufficient inbound innovation, firms risk depleting their internal knowledge base, making it difficult to maintain competitiveness and innovation capacity. Furthermore, overreliance on outbound innovation increases risks associated with intellectual property protection and raises coordination costs, reducing responsiveness to market demands and entrepreneurial agility (Capurro *et al.*, 2021). Thus, imbalances in these open innovation dimensions disrupt the alignment of knowledge acquisition and dissemination processes, leading to inefficiencies that hinder entrepreneurial performance.

H12. An incongruent increase of inbound and outbound open innovation in either direction would decrease entrepreneurial performance.

The hypothesized model is illustrated in Figure 1.

3. Methods

3.1 Data collection and sample

Vietnam, a rapidly developing country in Southeast Asia (Nguyen *et al.*, 2024), offers a compelling context for studying the dynamics of AI innovation, open innovation, and entrepreneurial performance due to its vibrant entrepreneurial ecosystem and increasing focus on technology-driven ventures (Duong, 2024). The country's entrepreneurial landscape is marked by high business ownership rates, which stand at 19.6%, and a robust startup ecosystem comprising approximately 4,000 ventures as of 2024 (Đoàn Xuân, 2024). Despite these advancements, challenges persist, as business survival rates remain low for three years (Lê Thị Anh and Dương Thùy, 2024). Vietnam's startup ecosystem, consisting of approximately 4,000 innovative ventures as of 2024, has demonstrated significant growth, particularly in high-tech sectors (Đoàn Xuân, 2024). Globally, Vietnam ranks 56th out of 100 countries in the Global Startup Ecosystem Index, with Hanoi and Ho Chi Minh City listed among the top 200 innovative startup cities worldwide. Key high-tech parks, including Hoa Lac Hi-Tech Park, Saigon Hi-Tech Park, Da Nang Hi-Tech Park, and the recently established Ha Nam Hi-Tech Park, serve as crucial centers for fostering technological advancement and innovation. These hubs provide a valuable context for examining AI innovation and its role in innovation niches, open innovation and entrepreneurial performance.

The study targeted new ventures, which were established within eight years of data, as defined by Wu *et al.* (2022) and employed at least four employees. A total of 1,100 founders of

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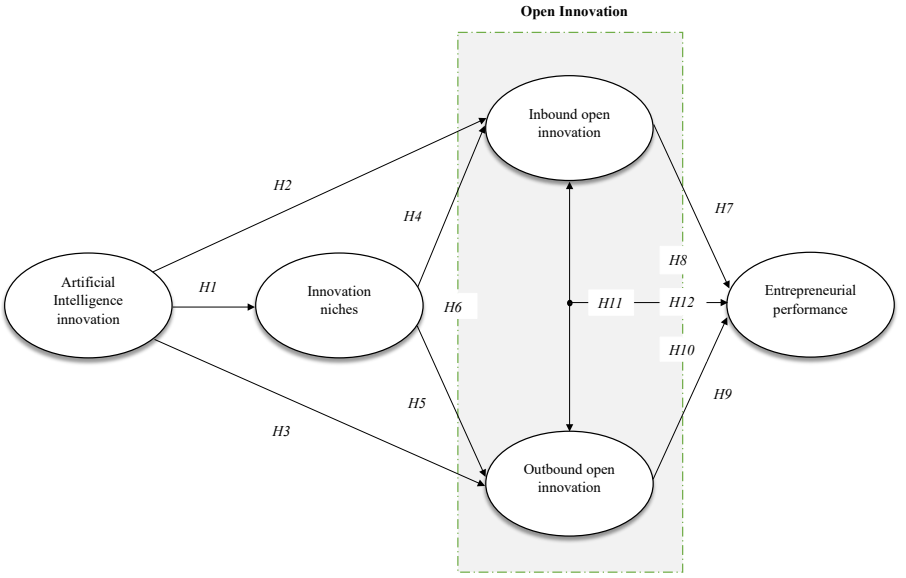


Figure 1. Conceptual framework. Source: Author's proposition

new ventures were identified and contacted for participation in the study. The sample was drawn from ventures located in high-tech parks and other innovation-driven firms across Vietnam to ensure relevance to the research focus. These firms are often early adopters of advanced technologies, including AI, which they use to improve operational efficiency, develop innovative products, and enhance service delivery (Han *et al.*, 2024). Additionally, these firms emphasize innovation development activities, such as research and development (Jin *et al.*, 2022), which aligns closely with the study's objective of examining the relationship between AI innovation, open innovation, and entrepreneurial performance. To ensure that the sample was representative of the broader Vietnamese entrepreneurial ecosystem, we employed stratified sampling across diverse industries and firm sizes. The sample includes ventures operating in various sectors such as information technology, manufacturing, healthcare, and education, which together reflect the sectoral distribution of Vietnam's broader startup landscape. Furthermore, demographic profiles (e.g. firm age, size, and geographical distribution) were compared to national-level reports on Vietnamese entrepreneurial ventures, confirming a close alignment in terms of age structure (majority under 8 years), employment size (ranging from 4 to 50 employees), and regional concentration in major innovation hubs. This increases the external validity of our findings and suggests the sample is broadly reflective of the high-tech entrepreneurial population in Vietnam. Data collection involved distributing personalized packets containing a cover letter, survey instrument, and a pre-paid return envelope to the selected participants. To enhance participation, follow-up emails and weekly reminders were sent to non-respondents, and phone calls were made to emphasize the study's importance. The data collection process spanned three months, from September 2024 to November 2024. Out of the 1,100 surveys distributed, 26 were returned as undeliverable due to incorrect contact details or the unavailability of the recipients. Of the 1,074 deliverable surveys, 340 responses were received, yielding a response rate of 31.7%. After excluding 11 incomplete responses, the final dataset comprised 329 usable responses, ensuring robust data quality for subsequent analysis. This final sample is consistent with prior studies using similar sampling frames in emerging market contexts and provides sufficient statistical power for the planned multivariate analyses (Xiao *et al.*, 2025). Table 1 presents the sample's characteristics.

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Table 1. Sample characteristics

Variables	Sub-category	Frequency	%
Founders' gender	Male	261	81.5
	Female	61	18.5
Founders' age	<30 years old	139	42.2
	30–40 years old	152	46.2
	>40 years old	38	11.6
Industry	Manufacturing	180	54.7
	Services	149	45.3
Venture age	<2 years	59	17.9
	2–4 years	121	36.8
	4–6 years	102	31.0
	6–8 years	47	14.3
Venture size	0–9 employees	151	45.9
	10–20 employees	114	34.7
	21–30 employees	38	11.6
	>30 employees	26	7.9

Note(s): $N = 329$
Source(s): Author's elaborations based on the research data

3.2 Scales

The measurement scales used in this study were adapted from established sources to ensure validity and reliability. AI innovation was assessed using four items derived from the research of Ku and Chen (2024). Innovation niches were measured with five items based on the study of Xie and Wang (2021). Both inbound open innovation and outbound open innovation were assessed with five items each, adapted from Wang *et al.* (2023). Entrepreneurial performance was evaluated using a nine-item scale developed from the studies of Huang *et al.* (2014) and Li and Atuahene-Gima (2001). All items were rated on a 5-point Likert scale, ranging from 1 (strongly disagree) to 5 (strongly agree). Further details on these scales are presented in Table 2.

4. Results

4.1 Reliability and validity test

Confirmatory factor analyses (CFAs) and α were conducted to assess the constructs' reliability and validity. Initially, items OOI5 (0.893) and EP4 (0.905) were removed because the "α if item deleted" values for the respective constructs were higher than their original α values (0.763 for OOI and 0.904 for EP) (Nunnally and Bernstein, 1994). Similarly, items INN1 (0.270) and INN5 (0.377) were excluded due to factor loadings below the acceptable threshold of 0.50 (Brown, 2006). After eliminating the unsatisfactory items, The results confirmed acceptable model fit indices for the hypothesized five-factor model: $\chi^2 = 416.566$, df = 232, GFI = 0.904, CFI = 0.964, TLI = 0.958, GFI = 0.904, and RMSEA = 0.049 (Hair *et al.*, 2021). Moreover, results in Table 2 indicate that all constructs demonstrated α values exceeding 0.70, satisfying the threshold for internal consistency. Additionally, CR values for all constructs were above 0.70, indicating strong construct reliability. Moreover, convergent validity was assessed using each construct's AVE. The AVE values exceeded 0.50 for all constructs, meeting the threshold (Anderson and Gerbing, 1988). The square root of the AVE values, shown along the diagonal of Table 3, was greater than the correlations between constructs, indicating satisfactory discriminant validity (Chin, 1998).

4.2 Common method bias test

Harman's single-factor test was initially conducted to examine potential common method bias. The unrotated first factor accounted for 40.045% of the variance, which is below the critical

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Table 2. Internal consistency and converged validity

Constructs/scales	Codes	Measures	Sources	Loading	α	CR	AVE
AI innovation	AII1	The development and experimentation of artificial intelligence involves our organizational strategy	Ku and Chen (2024)	0.780	0.877	0.878	0.643
	AII2	The efficiency of artificial intelligence is improved with our strategic intelligence		0.813			
	AII3	An artificial intelligence system supports our organizational strategies by reviewing processes where intelligent strategy can be implemented		0.823			
	AII4	Artificial intelligence is essential for our enhanced organizational performance		0.790			
Innovation niches	INN1	Compared with other firms, our company has a larger scale of operation	Xie and Wang (2021)	–	0.806	0.814	0.596
	INN2	Compared with other firms, our company has certain advantages in technical resources		0.803			
	INN3	Compared with other firms, our company has certain advantages in capital resources		0.864			
	INN4	Compared with other firms, our company has certain advantages in market resources		0.631			
	INN5	Compared with other firms, our company has certain advantages in human resources		–			
Inbound open innovation	IOI1	We often acquire knowledge or technology from external sources	Wang *et al.* (2023)	0.808	0.906	0.905	0.655
	IOI2	We regularly look for external ideas that might create value for us		0.793			
	IOI3	We have well-established systems for finding and acquiring external knowledge or technology		0.870			
	IOI4	We reach out to external parties to obtain more useful knowledge or technology		0.785			
	IOI5	We forge closer ties with external parties and rely on them for innovative activities		0.789			

(*continued*)

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Table 2. Continued

Constructs/scales	Codes	Measures	Sources	Loading	α	CR	AVE
Outbound open innovation	OOI1	We actively manage the flow of knowledge or technology to the outside world	Wang *et al.* (2023)	0.832	0.893	0.894	0.678
	OOI2	We regard the sale of knowledge or technology on the market as a formal business		0.829			
	OOI3	We have a dedicated department to commercialize intellectual assets		0.799			
	OOI4	We welcome the purchase and use of our intellectual property or technology by others		0.832			
	OOI5	We rarely work with external organizations to develop knowledge or technology		–			
Entrepreneurial performance	EP1	Our firm's return on assets is much better than that of competitors	Huang *et al.* (2014) and Li and Atuahene-Gima (2001)	0.709	0.905	0.904	0.542
	EP2	Our firm's return on sales is much better than that of competitors		0.789			
	EP3	Our firm's return on equity is much better than that of competitors		0.750			
	EP4	Our firm's cash flow is much better than that of competitors		0.727			
	EP5	Our firm's gross profit margin is much better than that of competitors		0.752			
	EP6	Our firm's sales growth is much better than that of competitors		0.668			
	EP7	Our firm's market share growth is much better than that of competitors		–			
	EP8	Our firm's profit growth is much better than that of competitors		0.709			
	EP9	Our firm's employment growth is much better than that of competitors		0.778			

Note(s): $N = 329$, α: Cronbach's alpha. CR = Composite reliability; AVE = Average variance extracted
Source(s): Author's elaborations based on the research data

threshold of 50%, indicating that common method bias is not a significant concern (Harman, 1976). Additionally, CFAs were employed to compare the hypothesized five-factor model with alternative models. The five-factor model exhibited superior fit indices (e.g. CFI = 0.964, RMSEA = 0.049) compared to the one-factor model (CFI = 0.643, RMSEA = 0.143), with significant differences in model fit ($\Delta\chi^2 = 1525.458$, $\Delta df = 20$, $p < 0.001$) (see Table 4). These results confirm that common method bias does not pose a significant threat to the validity of the findings.

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Table 3. Descriptive statistics and correlations

Variables	Mean	S.D.	1	2	3	4	5	6	7	8	9	10
1. Entrepreneur' age	1.693	0.667	–									
2. Entrepreneur' gender	1.185	0.389	−0.062	–								
3. Educational background	2.100	0.517	−0.025	0.119*	–							
4. Established years	2.416	0.943	−0.009	−0.003	−0.017	–						
5. Venture size	1.815	0.927	0.011	−0.006	−0.076	0.029	–					
6. OOI	3.660	0.677	−0.013	0.023	−0.026	0.050	−0.041	*0.823*				
7. AII	3.686	0.653	−0.035	0.008	−0.037	0.022	−0.042	0.803**	*0.802*			
8. INN	2.980	0.918	0.003	0.204**	0.036	−0.050	−0.043	0.184**	0.193**	*0.772*		
9. IOI	3.633	0.682	0.008	0.029	−0.004	0.017	−0.093	0.775**	0.748**	0.215**	*0.810*	
10. EP	3.536	0.748	0.076	−0.058	−0.102	0.014	0.053	0.469**	0.478**	0.174**	0.511**	*0.736*

Note(s): $N = 329$. $^{*}p < 0.05$, $^{**}p < 0.01$. The italic scores along the diagonal represent the square root of the AVE

Source(s): Author's elaborations based on the research data

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Table 4. Confirmatory factor analyses

Models		χ^2	df	GFI	CFI	TLI	NFI	RMSEA	$\Delta\ \chi^2$	Δ df
Five-factor model	AII, INN, IOI, OOI, EP	416.566	232	0.904	0.964	0.958	0.924	0.049	–	–
Four-factor model	AII + INN, IOI, OOI, EP	926.645	246	0.798	0.869	0.853	0.931	0.092	510.079***	14
Three-factor model	AII + INN + IOI, OOI, EP	1059.353	249	0.771	0.844	0.827	0.806	0.100	642.787***	17
Two-factor model	AII + INN + IOI + OOI, EP	1121.197	251	0.762	0.832	0.816	0.795	0.103	704.631***	19
One-factor model	AII + INN + IOI + OOI + EP	1942.024	252	0.581	0.674	0.643	0.645	0.143	1525.458***	20

Note(s): $N = 329$. * $p < 0.05$. Reference model: Five unconstrained constructs; *** $p < 0.001$

Source(s): Author's elaborations based on the research data

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4.3 Hypothesis testing

Hierarchical multiple regression, PROCESS macro and PRRSA were employed to evaluate the proposed hypotheses. PRRSA is a robust method for examining both linear and curvilinear relationships, allowing for an in-depth analysis of congruence and discrepancies between constructs. The regression results are presented in Table 5.

The findings confirm that AI innovation significantly enhances innovation niches ($\beta = 0.350$, $p < 0.001$), inbound open innovation ($\beta = 0.747$, $p < 0.001$), and outbound open innovation ($\beta = 0.803$, $p < 0.001$). These results provide strong support for H1, H2, and H3. Additionally, innovation niches positively influence both inbound open innovation ($\beta = 0.097$, $p < 0.05$) and outbound open innovation ($\beta = 0.084$, $p < 0.05$), supporting H4 and H5. The mediation analysis (Table 6) further reveals that innovation niches mediate the relationship between AI innovation and inbound open innovation ($\beta = 0.035$, 95% CI [0.009, 0.068]), as well as outbound open innovation ($\beta = 0.029$, 95% CI [0.007, 0.057]). These results confirm H6. The results also demonstrate that inbound open innovation positively impacts entrepreneurial performance ($\beta = 0.465$, $p < 0.001$), as does outbound open innovation ($\beta = 0.422$, $p < 0.001$), providing support for H7 and H9. Furthermore, mediation analyses indicate that inbound open innovation mediates the relationship between AI innovation and entrepreneurial performance ($\beta = 0.298$, 95% CI [0.176, 0.427]) as well as the relationship between innovation niches and entrepreneurial performance ($\beta = 0.117$, 95% CI [0.057, 0.189]). H8 is thus supported. Moreover, outbound open innovation exhibits similar mediating effects on the impacts of AI innovation and innovation niches on entrepreneurial performance ($\beta = 0.221$, 95% CI [0.093, 0.348]; $\beta = 0.049$, 95% CI [0.008, 0.110]), supporting for H10.

Response surface analysis was conducted to evaluate the effect of congruence between inbound open innovation and outbound open innovation on entrepreneurial performance. First, we analyzed the line of perfect agreement (IOI = OOI), which reflects a situation where a firm balances its inbound and outbound open innovation efforts equally. Along this line, the slope ($\tilde{a}1 = 0.620$, $p < 0.001$) was significantly positive, indicating that as both IOI and OOI increase in tandem, EP also increases. In practical terms, this suggests that firms engaging in both high levels of inbound and outbound open innovation simultaneously tend to achieve superior performance outcomes. The curvature along this line ($\tilde{a}2 = 0.040$, $p > 0.05$) was not significant, indicating that the increase in performance is linear rather than curvilinear. This means that the performance benefits of alignment do not accelerate or decelerate at higher levels; they simply rise steadily. These findings are visually represented in Figure 2, where the highest levels of EP appear in the back-right corner of the 3D surface plot—where both IOI and OOI are maximized and aligned. Conversely, the lowest EP values are observed in the front-left corner, where both IOI and OOI are low. These results provide strong support for H11 and emphasize that a balanced and high-level open innovation strategy yields better outcomes than imbalanced or low-level strategies.

Second, to test H12, we evaluated the line of incongruence (IOI ≠ OOI), which captures misalignment scenarios where one form of open innovation outweighs the other. Along this line, the slope ($\tilde{a}3 = -0.310$, $p < 0.05$) was significantly negative, demonstrating that as the disparity between IOI and OOI widens—regardless of which is higher—EP tends to decline. This suggests that firms disproportionately emphasizing either inbound or outbound open innovation risk underperformance due to inefficiencies or underutilized capabilities. However, the curvature along this line ($\tilde{a}4 = 0.090$, $p > 0.05$) was not significant, which indicates that the decline in performance is relatively constant and does not intensify at greater degrees of misalignment. Figure 2 visually reinforces this interpretation: performance steadily drops as the surface moves away from the IOI = OOI line toward the misalignment zone. These results validate H12 and demonstrate that misalignment between open innovation modes has a consistently negative but not exponentially worsening effect on performance. These analyses affirm that alignment between IOI and OOI is linearly beneficial, while imbalance yields proportionate but not extreme performance penalties. By clarifying the interpretation of slopes

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Table 5. Polynomial regression and response surface analysis

Variables	Innovation niches		Inbound open innovation		Outbound open innovation		Entrepreneurial performance		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9
Constant	2.987***	1.636***	3.683***	0.514*	3.685***	0.339	3.691***	1.425***	0.126
Entrepreneur' age	0.034	0.046	0.011	0.033	−0.011	0.013	0.079	0.076	0.073
Entrepreneur' gender	0.269***	0.263	0.055	0.016	0.046	0.009	−0.082	−0.114	−0.129
Educational background	−0.027	−0.008	−0.019	0.023	−0.042	0.003	−0.132	−0.116	−0.112
Established years	−0.026	−0.032	0.014	0.005	0.036	0.026	0.009	−0.004	−0.003
Venture size	−0.004	0.007	−0.070	−0.045	−0.033	−0.007	0.036	0.072	0.070
AII		0.350***		0.747***		0.803***			
INN				0.097*		0.084*			
IOI								0.422***	0.465***
OOI								0.194*	0.151*
IOI*IOI									0.160
IOI*OOI									−0.028
OOI*OOI									−0.096
R^2	0.024	0.132	0.010	0.574	0.006	0.652	0.020	0.300	0.313
Adj-R^2	0.009	0.116	−0.005	0.565	−0.009	0.645	0.005	0.284	0.291
ΔF	3.111**	8.154***	0.672	61.909***	0.387	86.028***	0.020	19.626***	14.464***
Congruent line									
Slope ($\tilde{a}_1$)									0.620***
Curvature ($\tilde{a}_2$)									0.040
Incongruent line									
Slope ($\tilde{a}_3$)									0.310*
Curvature ($\tilde{a}_4$)									0.090

Note(s): $N = 329$. *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$

Source(s): Author's elaborations based on the research data

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Table 6. Mediation analyses

Indirect effects							β	SE	Bootstrap 95% CIs LLCI	ULCI
AII	→	INN	→	IOI			0.035	0.015	0.009	0.068
AII	→	INN	→	OOI			0.029	0.013	0.007	0.057
AII	→	IOI	→	EP			0.298	0.063	0.176	0.427
INN	→	IOI	→	EP			0.117	0.034	0.057	0.189
AII	→	OOI	→	EP			0.221	0.064	0.093	0.348
INN	→	OOI	→	EP			0.049	0.028	0.001	0.110
AII	→	INN	→	IOI	→	EP	0.011	0.006	0.003	0.025
AII	→	INN	→	OOI	→	EP	0.006	0.004	0.001	0.014

Note(s): $N = 329$; SE: Standard errors

Source(s): Author's elaborations based on the research data

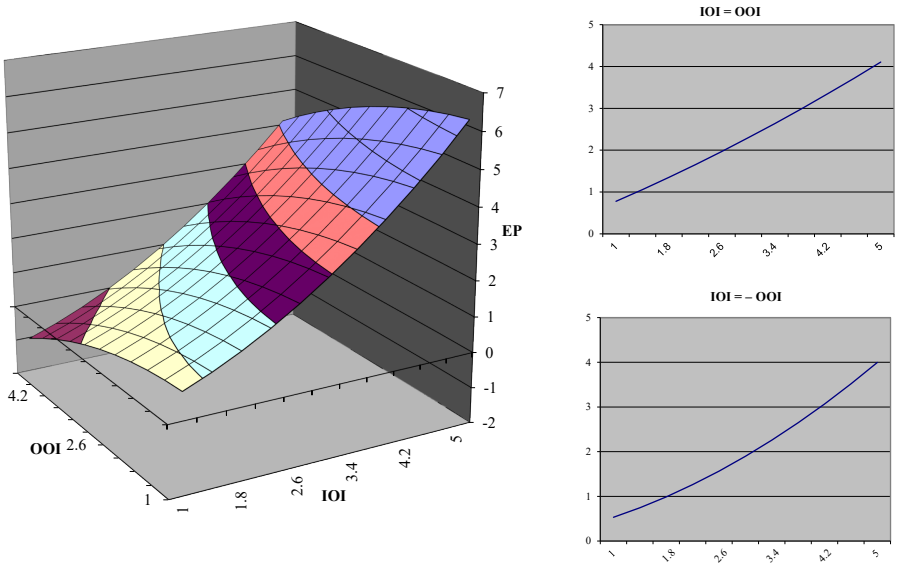


Figure 2. Response surface analysis of exploratory and exploitative innovations. Source: Author's elaborations based on the research data

(direction of change) and curvatures (rate of change), this response surface analysis offers nuanced insights into how symmetry and asymmetry in innovation strategy affect entrepreneurial outcomes.

4.4 Robustness tests

To reinforce the validity of the main findings and address concerns regarding model robustness and potential omitted variables (e.g. firm age, size, and industry), a robustness check was conducted using partial least squares structural equation modeling (PLS-SEM). As shown in Table 7 and Figure 3, the structural model reaffirmed the original results, confirming the significant positive effects of AI innovation on innovation niches ($\beta = 0.194$, $p = 0.001$), inbound open innovation ($\beta = 0.736$, $p < 0.001$), and outbound open innovation ($\beta = 0.797$,

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Table 7. Robustness tests

Paths			β	S.D.	*t*-value	*p*	f^2	VIF
AI innovation	→	Innovation niches	0.194	0.061	3.193	0.001	0.039	1.000
AI innovation	→	Inbound open innovation	0.736	0.036	20.693	0.000	1.206	1.039
AI innovation	→	Outbound open innovation	0.797	0.028	28.374	0.000	1.732	1.039
Innovation niches	→	Inbound open innovation	0.072	0.035	2.080	0.038	0.012	1.039
Innovation niches	→	Outbound open innovation	0.031	0.031	1.005	0.315	0.003	1.039
Inbound open innovation	→	Entrepreneurial performance	0.356	0.078	4.572	0.000	0.069	2.535
Outbound open innovation	→	Entrepreneurial performance	0.191	0.074	2.580	0.010	0.020	2.526
Entrepreneurs' age	→	Entrepreneurial performance	0.083	0.049	1.691	0.091	0.010	1.002
Established years	→	Entrepreneurial performance	0.014	0.049	0.295	0.768	0.000	1.005
Venture size	→	Entrepreneurial performance	0.096	0.047	2.020	0.043	0.012	1.013
Industry	→	Entrepreneurial performance	−0.103	0.097	1.065	0.287	0.004	1.012

Note(s): $N = 329$

Source(s): Author's elaborations based on the research data

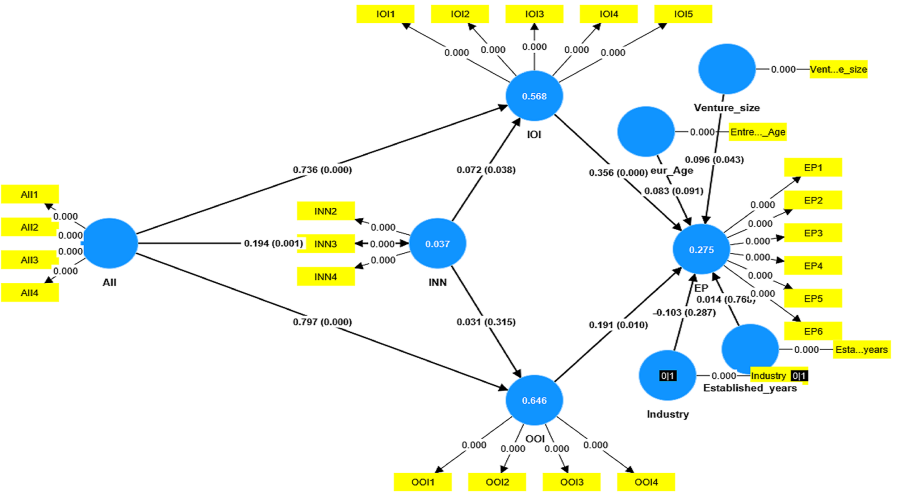


Figure 3. Robustness test with PLS-SEM. Source: Author's elaborations based on the research data

$p < 0.001$). Likewise, both inbound ($\beta = 0.356$, $p < 0.001$) and outbound open innovation ($\beta = 0.191$, $p = 0.010$) significantly enhanced entrepreneurial performance.

Additionally, the influence of control variables was analyzed. Entrepreneurial performance was positively associated with entrepreneur age ($\beta = 0.083$, $p = 0.091$) and venture size ($\beta = 0.096$, $p = 0.047$), while industry and firm age showed no significant effects. The inclusion of these controls further increases confidence in the robustness of the reported

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relationships. Variance inflation factor (VIF) values remained below the threshold of 3.3 for all constructs (O'brien, 2007), confirming the absence of multicollinearity concerns. Collectively, these robustness tests demonstrate that the proposed relationships are stable and consistent across alternative model specifications and when accounting for potential confounding factors.

5. Discussion

This study offers valuable insights into the interconnected roles of AI innovation, innovation niches, open innovation (both inbound and outbound), and entrepreneurial performance. The findings emphasize the transformative impact of AI innovation in fostering a robust innovation ecosystem and enhancing entrepreneurial success.

First, AI innovation is shown to be a critical driver for both innovation niches and open innovation activities. By enabling firms to harness external knowledge and disseminate internal knowledge effectively, AI plays a key role in fostering inbound and outbound open innovation. Firms leveraging AI technologies are better equipped to process vast amounts of data, identify patterns, and make informed strategic decisions. These capabilities enable them to strengthen connections within innovation ecosystems and optimize innovation processes. Furthermore, the mediating role of innovation niches demonstrates how AI innovation creates value indirectly by enhancing firms' strategic positioning within their ecosystems. This finding aligns with the STS perspective (Münch *et al.*, 2022), which shows the importance of integrating technological advancements with organizational processes to achieve innovative outcomes (Ciriello *et al.*, 2024; Sahoo *et al.*, 2024).

Second, the study highlights the importance of innovation niches as enablers of open innovation. Our study shows that innovation niches serve as a foundation for acquiring external knowledge (inbound open innovation) and transferring internal knowledge to external entities (outbound open innovation). However, the relatively weaker effect of innovation niches on outbound open innovation, compared to their influence on inbound innovation, suggests that while niches provide resource advantages, these may not always translate directly into outward knowledge flows. This points to a possible decoupling between structural advantages and external commercialization capabilities (Gentile-Lüdecke *et al.*, 2020), particularly in emerging ecosystems. This finding is consistent with some prior studies (Xie and Wang, 2021; Lopolito *et al.*, 2013), which emphasizes that innovation niches allow firms to maintain a competitive edge by capitalizing on their distinctive capabilities while engaging with broader innovation networks.

The findings also reveal the significant contributions of inbound and outbound open innovation to entrepreneurial performance. Inbound open innovation enhances a firm's ability to absorb and utilize external knowledge, enabling it to adapt to market changes and innovate effectively (Liao *et al.*, 2020; Abazi-Alili *et al.*, 2024; Moretti and Biancardi, 2020). Outbound open innovation, in contrast, supports the commercialization and dissemination of internal knowledge, creating new revenue streams and strengthening external partnerships (Ode and Ayavoo, 2020; Audretsch *et al.*, 2023). Together, these complementary dimensions of open innovation highlight the need for firms to adopt a balanced approach (Wang *et al.*, 2023), integrating both inbound and outbound activities to achieve optimal entrepreneurial performance. Indeed, the response surface analysis provides valuable insights into the alignment and misalignment between inbound and outbound open innovation. When these dimensions are aligned and increase simultaneously, entrepreneurial performance improves significantly, suggesting a synergistic effect. This alignment enables firms to leverage their open innovation activities more effectively, leading to enhanced performance. Conversely, misalignment—where one dimension exceeds the other—diminishes entrepreneurial performance, emphasizing the importance of balance. Notably, the results show that discrepancies between inbound and outbound open innovation consistently harm performance, regardless of the degree of misalignment. These findings stress the critical

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need for firms to manage congruence between inbound and outbound open innovation to sustain their entrepreneurial performance.

While these findings offer theoretically and practically significant insights, they should be interpreted within the context of Vietnam's entrepreneurial ecosystem. Vietnam represents an emerging economy with a government-supported innovation infrastructure and evolving AI maturity (Mai *et al.*, 2024). Consequently, the generalizability of the findings may be limited in more mature or institutionally different settings. In contrast to Western ecosystems where firms may possess greater absorptive capacity, legal protections, and commercialization networks, Vietnamese ventures may experience higher reliance on government policies and informal networks (Đoàn Xuân, 2024). This context-specificity should be considered when applying the results internationally. Furthermore, these results contribute to the global innovation debate by providing empirical evidence from a non-Western, high-growth entrepreneurial economy. In mature ecosystems like Europe or North America, where open innovation is often institutionalized, the performance effects of inbound-outbound congruence may manifest differently. This study adds to the growing body of research that highlights the importance of geographical and developmental context in shaping innovation dynamics and reinforces calls for more diverse empirical samples in innovation and entrepreneurship research.

6. Conclusions

6.1 Theoretical contributions

This study makes notable theoretical contributions to the fields of AI innovation, open innovation, and entrepreneurial performance by integrating the STS perspective and examining the roles of innovation niches. Our findings provide fresh insights into how AI innovation impacts innovation ecosystems and entrepreneurial outcomes, addressing gaps in the current literature.

First, this study advances the open innovation literature by addressing a specific gap in the understanding of congruence effects between inbound and outbound open innovation. While prior studies have demonstrated the role of AI in improving operational efficiency and decision-making (Kumar *et al.*, 2024; Giachino *et al.*, 2024), no any studies have explored their alignment or misalignment. Using PRRSA—a rarely applied method in this domain—this study reveals that alignment generates additive benefits, whereas misalignment imposes consistent performance penalties. This nuanced understanding provides empirical validation for the theoretical proposition that innovation strategy alignment enhances organizational fit and outcomes (Broekhuizen *et al.*, 2023; Prasetyo *et al.*, 2025).

Second, the research contributes to the AI innovation discourse by addressing calls for more entrepreneurial-context studies (Giuggioli and Pellegrini, 2022; Mariani and Dwivedi, 2024). Whereas existing research has predominantly examined AI's role in mature firms and operational efficiency (Sahoo *et al.*, 2024; Broekhuizen *et al.*, 2023), this study uniquely focuses on new ventures, highlighting how AI innovation acts as an enabler of knowledge acquisition, collaboration, and strategic agility under conditions of uncertainty and constraint. This comparative lens underscores the distinctive value of AI in resource-scarce environments—an area underrepresented in the literature (Kraus *et al.*, 2022).

Third, we build on and extend the concept of innovation niches by positioning them as dynamic mediators that connect technological inputs (AI innovation) with open innovation processes. While Lopolito *et al.* (2013) and Xie and Wang (2021) discussed niches in ecosystem positioning, few studies have empirically tested their bridging role between digital technologies and innovation strategies. Our findings support this intermediary function and provide evidence that innovation niches amplify the influence of AI innovation on both inbound and outbound open innovation. This finding aligns with the resource orchestration perspective (Schepis *et al.*, 2021), emphasizing how firms can structure and leverage unique resource configurations to drive innovation.

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Fourth, the integration of the STS perspective into this study enriches its theoretical depth. While STS theory has been used to describe macro-level technological transformations (Thomas, 2024), our application focuses on the micro-foundations within new ventures, showing how social (e.g. collaborative culture) and technical (e.g. AI platforms) subsystems must co-evolve to enable innovation. This insight addresses a previously superficial treatment of STS in entrepreneurial contexts and underscores the dynamic interplay between organizational culture and digital tools.

Finally, this research contributes a comparative theoretical roadmap for future studies by identifying the contextual boundary conditions, such as national ecosystem maturity and digital infrastructure, that shape the efficacy of innovation strategies. By positioning Vietnam's startup ecosystem as a contrast case to Western innovation environments, this study highlights how institutional conditions moderate the effectiveness of AI-enabled innovation strategies (Chalmers *et al.*, 2021).

6.2 Practical and managerial implications

The findings of this study provide several actionable insights for entrepreneurs, managers, and policymakers, especially in the context of new ventures navigating dynamic and competitive environments. These implications are particularly relevant for startups seeking to strategically leverage AI innovation within open innovation ecosystems.

Entrepreneurs and managers should focus on leveraging AI not merely as a technological enhancement but as a strategic enabler of both inbound and outbound open innovation. The study demonstrates that AI innovation significantly contributes to the development of innovation niches while enhancing firms' capacities to absorb external knowledge and disseminate internal knowledge. For instance, an AI-driven fintech startup in Vietnam could use natural language processing to analyze user feedback (inbound knowledge), while simultaneously offering its AI models as APIs to other service providers (outbound knowledge), thereby creating mutual value and extending its innovation reach. This kind of dual-use capability supports agile adaptation to rapidly evolving markets, improves decision-making, and enhances overall entrepreneurial performance. Managers should embed AI tools within broader innovation strategies to ensure their alignment with business objectives, organizational processes, and collaborative goals.

Equally important is the need for firms to align their inbound and outbound open innovation activities. The study's results emphasize that alignment between these two dimensions produces a synergistic effect that significantly improves entrepreneurial outcomes, whereas imbalance—where one is emphasized over the other—tends to reduce performance. A practical illustration can be seen in the case of an AI healthcare startup that prioritizes absorbing clinical data from hospitals and public datasets (inbound), but neglects to share its internally developed diagnostic algorithms or insights (outbound). Without contributing back to the ecosystem, the firm may struggle with visibility, partnerships, or market entry. Achieving balance—by co-developing solutions with hospitals or publishing validated models—can foster greater innovation traction and credibility. Managers are thus encouraged to continuously monitor and recalibrate their open innovation efforts, ensuring that both acquisition and dissemination are strategically aligned and adequately resourced. AI-powered analytics can play a vital role in providing real-time feedback on these dynamics, identifying gaps, and informing tactical adjustments.

Innovation niches are also shown to mediate the relationship between AI innovation and open innovation. For new ventures, cultivating a strong innovation niche means identifying their unique combinations of technological capabilities, market positioning, and resource configurations. Startups in robotics or AI-enhanced agriculture, for example, can establish niche dominance by participating in government-sponsored pilot programs or partnering with research institutions, allowing them to test and validate new technologies in real-world settings. These engagements not only refine internal competencies but also strengthen external

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trust and collaboration, supporting both inbound learning and outbound sharing. Policymakers and incubators can facilitate this process by offering targeted support, such as technical training, R&D infrastructure, and access to collaborative platforms that are tailored to the needs of technology-focused startups.

Moreover, AI innovation plays a transformative role in enhancing startups' decision-making capabilities. AI enables ventures to make more informed, agile, and forward-looking decisions by processing large-scale data inputs, identifying trends, and forecasting opportunities. These capabilities are particularly critical in emerging ecosystems like Vietnam's, where uncertainty and resource limitations are common. For example, a new AI startup could use predictive analytics not only to discover unmet market needs but also to identify optimal partners for inbound knowledge exchange or to select the most promising channels for commercializing outbound innovations. This allows firms to act decisively and reduce trial-and-error, accelerating both growth and resilience.

To make these insights more actionable for entrepreneurs, this study implicitly outlines a strategic roadmap. First, startups should begin by mapping their current innovation activities to diagnose imbalances or blind spots. Then, they should integrate AI tools to support both external knowledge sourcing and internal knowledge sharing. Identifying and nurturing a defensible innovation niche—through partnerships, IP, or platform positioning—can reinforce these efforts. Finally, firms should institutionalize a continuous monitoring and learning process, supported by AI-enabled dashboards and feedback loops, to ensure strategic coherence over time. This integrated approach will help new ventures harness the full potential of AI and open innovation in a way that is both context-sensitive and performance-oriented.

6.3 Limitations of the current study

This study has several limitations that present valuable opportunities for future research. First, the study focuses on new ventures in a specific geographical (Vietnam) and industry context (high-tech sector), which may limit the generalizability of the findings to other regions or sectors. Vietnam's entrepreneurial landscape is shaped by distinctive institutional conditions, including centralized innovation policies, state influence on resource allocation, and nascent regulatory frameworks for AI deployment. These factors may influence how AI innovation and open innovation processes unfold and interact. Future research could explore the applicability of the proposed framework across different national contexts (e.g. emerging versus developed economies) and industries (e.g. manufacturing, healthcare, or education) where AI maturity and open innovation practices differ significantly. Comparative cross-country analyses may help unpack the role of institutional and cultural specificities in shaping entrepreneurial innovation.

Second, the study employs a cross-sectional design, which restricts the ability to draw causal inferences. Although the model is theoretically grounded and supported by robust statistical methods (e.g. PRRSA and PLS-SEM), the use of cross-sectional data prohibits conclusions about temporal ordering or causality among constructs. The reliance on single-time-point data introduces potential biases, such as reverse causality and temporal misalignment between variables. To mitigate this, future studies should adopt longitudinal designs or experimental approaches to observe how AI innovation, innovation niches, and open innovation strategies co-evolve over time and impact entrepreneurial performance at different stages.

Third, although this research examines the alignment and misalignment of inbound and outbound open innovation, it does not consider the potential moderating effects of external environmental factors such as market dynamism, competitive intensity, or technological turbulence. Future studies could investigate how such contextual variables interact with open innovation congruence to amplify or dampen its effect on performance.

Finally, the study relies on self-reported data from single respondents per firm, which may introduce common method bias despite our statistical controls (e.g. CFA, Harman's one-factor

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test). While procedural and statistical techniques were used to minimize this bias, self-reported perceptual data are inherently limited in capturing objective performance. To enhance robustness, future research should consider multi-informant designs or combine perceptual and objective performance indicators (e.g. patent counts, funding records, or market share). Additionally, comparing the influence of different types of AI technologies (e.g. machine learning, generative AI, robotics) could reveal how varying technical affordances shape innovation mechanisms and outcomes.

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